

Intracranial Hypertension & Herniation

CLINICAL SIGNS OF HERNIATION

- **Motor decline:** Spontaneous GCS motor score decrease of ≥ 1 point.
- **Pupillary reactivity:** Decrease in pupillary reactivity (Neurological Pupil Index, NPi < 3).
- **Asymmetry:** New pupillary asymmetry or unilateral dilation (ipsilateral mydriasis).
- **Focal deficit:** New focal motor deficit or abnormal posturing (decorticate / decerebrate).
- **Cushing's Triad (Late Sign):** Systolic hypertension, bradycardia, and irregular respirations. ***Cushing triad is a LATE sign of brainstem compression.***

Syndrome	Anatomical Substrate	Clinical Exam & Diagnostic Trap
Uncal (Lateral)	Medial temporal lobe (uncus) pushed over tentorial edge	Ipsilateral sluggish/dilated pupil (CN III compressed), contralateral hemiparesis. Kernohan's Notch causes false-localizing ipsilateral hemiparesis.
Central (Axial)	Downward diencephalic and midbrain displacement	Progressive stupor, midpoint fixed pupils, decorticate to decerebrate posturing. Symmetrical signs often confused with metabolic encephalopathy.
Subfalcine (Cingulate)	Cingulate gyrus displaced under the falx cerebri	Often clinically silent, or presents with contralateral lower extremity weakness. ACA compression causes frontal/leg territory infarction.
Tonsillar (Downward)	Cerebellar tonsils forced through the foramen magnum	Cushing's triad, flaccid quadriplegia, respiratory arrest.

MANAGEMENT

GENERAL APPROACH

Tier 0: Fundamental Optimization

- Elevate HOB 30°; strict neutral midline neck alignment.
- Euvolemia (isotonic saline; avoid hypotonic D5W), normothermia (treat fevers), normocapnia ($p\text{CO}_2$ 35–40 mmHg).

Tier 1: Initial Interventions

- Continuous CSF drainage via EVD (typically set at 10–15 cmH_2O).
- Analgesia/sedation (propofol/fentanyl) to prevent coughing/agitation.

Tier 2: Hyperosmolar & Ventilatory Escalation

- **Mannitol 20%:** 1 g/kg IV bolus. **Hold if Osm > 320 or Gap ≥ 20 .**
- **HTS:** 3% (150–250 mL bolus) or 23.4% (30 mL rescue bolus; central access only). **Hold if Na > 155 or Cl > 115 .**
- **Controlled Hyperventilation:** Target $p\text{CO}_2$ 30–35 mmHg (temporary bridge).

Tier 3: Refractory & Salvage Measures

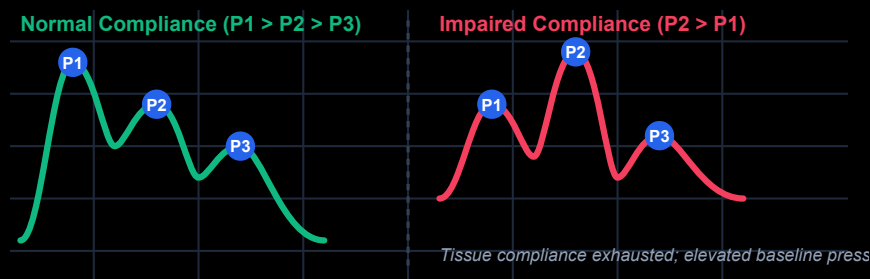
- High-dose barbiturate therapy (pentobarbital) titrated to burst suppression on EEG.

SURGICAL MANAGEMENT

Decompressive Surgery:

- **Malignant MCA (DHC):** Age ≤ 60 years, clinical decline (GCS decline ≥ 1 / pupil changes), infarct $\geq 50\%$ MCA territory, within 48h of onset (DECIMAL/DESTINY trials).
- **Cerebellar Stroke:** Suboccipital craniectomy for brainstem compression, 4th ventricle effacement, or obstructive hydrocephalus.

ICP WAVEFORM ANALYSIS



- **P1 (Percussion wave):** Arterial pulsation.
- **P2 (Tidal wave):** State of intracranial compliance (elastic reserve).
- **P3 (Dicrotic wave):** Venous pulsations.
- **Compliance States:**
 - **Normal Compliance:** $P1 > P2 > P3$ (elastic brain tissue easily cushions pulsations).
 - **Impaired Compliance / High ICP:** $P2 > P1$ (brain tissue reserve exhausted; high risk of herniation).

CPP = MAP - ICP

In patients with intracranial hypertension or mass effect, cerebral perfusion is highly pressure dependent, and cautious BP lowering is recommended.